The upshot is that no light reaches detector 2 at all, because the two beams heading toward it destructively interfere with each other—and the light reaching detector 1 is just as bright as the original laser beam that entered the first beam splitter in the lower left-hand corner of the experiment.

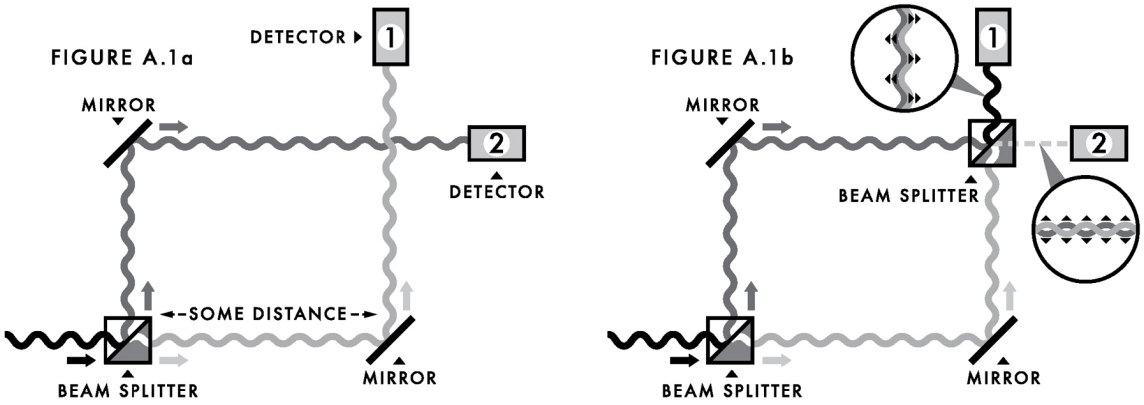


Figure A.1. Wheeler's delayed-choice experiment. (a) Without the second beam splitter, a single photon has a 50-50 chance of arriving at either detector. (b) With the second beam splitter, a single photon will interfere with itself, guaranteeing that it never arrives at detector 2.

So far, so good. Aside from the existence of the laser in the first place, everything we've described so far is just classical physics. Now let's get quantum. Make the laser beam really dim—as dim as possible, in fact, so we're just sending one photon at a time through this experiment. If we don't have the second beam splitter in the upper right-hand corner, this is still pretty simple. The photon is detected at either detector 1 or detector 2, and we can tell which path the photon took through the setup by seeing which detector it arrived at. And if we send lots of photons through, one at a time, about half will show up at each detector.

But, Wheeler says, things become much more complicated when we put the second beam splitter back into the experiment. If we do that, a photon will never arrive at detector 2 because it will interfere with itself, just as in the double-slit experiment. Send as many photons as you like through the experiment, one at a time, and they'll all show up at detector 1. This, Wheeler says, is because each photon took both paths and interfered with itself, preventing it from arriving at detector 2. By inserting the second beam splitter, says Wheeler, " we make the whole idea of following a single